1.4 Sequences & Series

Mathematics — OCR A-Level (H240) — Paper 1: Pure Mathematics

A *sequence* is an ordered list of numbers, each one called a *term*. A *series* is the sum produced by adding those terms together. The study of sequences and series lies at the centre of pure mathematics: arithmetic and geometric progressions appear in compound interest, population models, and physics problems; sigma notation compresses large sums into a single symbol; the binomial theorem extends polynomial expansion from exponents like $(1+x)^5$ into the infinite world of $(1+x)^{1/2}$ or $(1+x)^{-3}$. This topic also introduces the idea of a *limit* — the value a sequence approaches as more and more terms are produced — which foreshadows calculus.

1. Sequences — Notation and Definition

A sequence is a function whose domain is the positive integers (or sometimes $\{0, 1, 2, \ldots\}$). The nth term is denoted a_n or u_n; the whole sequence is written $(a_n)_{n\geq 1}$ or simply $a_1, a_2, a_3, \ldots$.

Sequence. An ordered list of numbers $a_1, a_2, a_3, \ldots$ indexed by a positive integer n. Each number a_n is a *term*; n is the *position* or *index* of that term. A sequence may be finite (with a last term) or infinite (continuing indefinitely).

Two Ways to Specify a Sequence

There are two standard ways to describe a sequence:

1. **Position-to-term formula** — an explicit expression $a_n = f(n)$ giving the nth term directly. Example: $a_n = 3n + 2$ gives $5, 8, 11, 14, \ldots$.
2. **Term-to-term (recurrence) formula** — a rule $a_{n+1} = g(a_n)$ expressing each term in terms of the preceding one(s), together with an initial term. Example: $a_{n+1} = a_n + 3$ with $a_1 = 5$ gives the same sequence $5, 8, 11, 14, \ldots$.

© RAG Learning · Not for redistribution

Recurrence relation. An equation $a_{n+1} = g(a_n)$ (or involving more earlier terms) that expresses each term of a sequence as a function of one or more preceding terms. Together with initial condition(s) this uniquely determines the sequence.

Exam tip — deducing a formula from the first few terms. If a sequence begins $2, 5, 8, 11, 14, \ldots$ and you spot a common difference $d = 3$, the explicit formula is $a_n = a_1 + (n-1)d = 2 + 3(n-1) = 3n - 1$. Always verify your guess on at least two terms (e.g. $n = 1$ gives 2 ✓; $n = 4$ gives 11 ✓). Do not just check the first term — many different formulae agree at $n = 1$.

Worked Example 1 — Converting between position-to-term and term-to-term

A sequence is defined by $a_{n+1} = 2a_n - 1$ with $a_1 = 2$. Compute $a_1, \ldots, a_5$ and find a position-to-term formula.

```
Compute terms:
   a₁ = 2
   a₂ = 2·2 − 1 = 3
   a₃ = 2·3 − 1 = 5
   a₄ = 2·5 − 1 = 9
   a₅ = 2·9 − 1 = 17.
Differences 1, 2, 4, 8 double → suggests 2ⁿ involvement.
Try aₙ = 2^(n−1) + 1:
   n=1: 1 + 1 = 2 ✓
   n=2: 2 + 1 = 3 ✓
   n=3: 4 + 1 = 5 ✓
   n=4: 8 + 1 = 9 ✓
∴ aₙ = 2^(n−1) + 1.      // verified explicit formula
```

2. Arithmetic Progressions (AP)

An arithmetic progression (AP) is a sequence in which each term is obtained from the previous one by adding a fixed constant, called the *common difference*.

Arithmetic progression. A sequence $a_1, a_2, a_3, \ldots$ such that $a_{n+1} - a_n = d$ for all n, where d is a fixed real constant (the *common difference*). Equivalently, the nth term is \[a_n = a + (n-1)d, \] where $a = a_1$ is the first term.

Sum of an AP

Let $S_n = a_1 + a_2 + \cdots + a_n$ denote the sum of the first n terms. Writing the sum forwards and backwards and adding term-by-term gives Gauss's formula:

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right] = \frac{n}{2}(a + l),$$

where $l = a_n = a + (n-1)d$ is the last (or nth) term. The second form is especially useful when both endpoints of the sum are known.

Exam tip — which AP sum formula to use. Use $S_n = \frac{n}{2}(a + l)$ when you already know the first *and* last term of the sum (e.g. "sum all integers from 1 to 100"). Use $S_n = \frac{n}{2}\left[2a + (n-1)d\right]$ when you know the first term and the common difference but not the last term. Both give the same answer — pick whichever avoids extra arithmetic.

Worked Example 2 — Sum of the first 100 positive integers

Compute $1 + 2 + 3 + \cdots + 100$.

```
1.  This is an AP:  a = 1,  d = 1,  n = 100,  l = 100.
2.  Using S_n = n/2 · (a + l):
3.     S₁₀₀ = 100/2 · (1 + 100) = 50 · 101 = 5050.
4.  Check via other formula:
5.     S₁₀₀ = 100/2 · [2·1 + 99·1] = 50 · 101 = 5050 ✓.

// Gauss famously produced this answer as a schoolboy by pairing
// 1+100, 2+99, … , 50+51 — fifty pairs each summing to 101.
```

Worked Example 3 — AP with known first and last terms

An AP has 20 terms, first term 5 and last term 100. Find the common difference and the sum.

```
Last term:  a_n = a + (n − 1)d.
   100 = 5 + 19d   ⇒   19d = 95   ⇒   d = 5.
Sum:  S_n = n/2 · (a + l)
   S₂₀ = 20/2 · (5 + 100) = 10 · 105 = 1050.
Quick sanity check — average term × number of terms:
   mean = (5 + 100)/2 = 52.5;   52.5 × 20 = 1050 ✓.
```

Applications of APs

- **Salary with fixed annual raise.** A worker earns £25 000 in year 1 with a £1 000 annual increase. Lifetime earnings over 30 years form an AP.
- **Pile of logs / cans.** A triangular stack with 1 log on top, 2 in the next row, $\ldots$, n at the base has total $\frac{n}{2}(1+n)$.
- **Walking / driving with constant change.** First day walk 5 km, each day walk 0.5 km more — total distance after n days is an AP sum.

3. Geometric Progressions (GP)

A geometric progression (GP) is a sequence in which each term is obtained by multiplying the previous one by a fixed non-zero constant called the *common ratio*.

Geometric progression. A sequence $a_1, a_2, a_3, \ldots$ such that $a_{n+1}/a_n = r$ for all n, where $r \neq 0$ is a fixed real constant (the *common ratio*). The nth term is \[a_n = a \cdot r^{n-1}, \] where $a = a_1$.

Sum of a GP

Let $S_n = a + ar + ar^2 + \cdots + ar^{n-1}$. Multiplying by r gives $rS_n = ar + ar^2 + \cdots + ar^n$. Subtracting:

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{a(1 - r^n)}{1 - r}, \qquad r \neq 1.$$

The two forms are algebraically identical; the first is neater when $|r| > 1$, the second when $|r| < 1$.

Sum to Infinity — Convergence

If $|r| < 1$ then $r^n \to 0$ as $n \to \infty$, and the infinite sum has a finite value:

$$S_\infty = \frac{a}{1 - r}, \qquad \text{valid only if } |r| < 1.$$

If $|r| \geq 1$ the partial sums do not settle to any single number — we say the series *diverges* and S_∞ is undefined.

Convergence of a GP. An infinite geometric series $a + ar + ar^2 + \ldots$ converges if and only if $|r| < 1$. When it converges, its sum is $S_\infty = a/(1 - r)$. When $|r| \geq 1$ (and $a \neq 0$) it diverges — the partial sums either grow without bound or oscillate.

Limit of a sequence. The sequence (a_n) has limit L (written $a_n \to L$ or $\lim_{n\to\infty} a_n = L$) if, informally, a_n can be made arbitrarily close to L by taking n sufficiently large. If no such L exists, the sequence diverges.

Exam tip — check convergence *before* summing. Every year OCR scripts lose marks by blindly applying $S_\infty = a/(1 - r)$ without verifying $|r| < 1$. If the ratio is $r = 2$ or $r = -3$, the infinite sum is undefined — writing down $a/(1 - r)$ anyway will score zero on that part. Always state the validity condition explicitly in your answer.

AP vs GP — at a glance

Feature	Arithmetic progression	Geometric progression
Recurrence	$a_{n+1} = a_n + d$	$a_{n+1} = r \cdot a_n$
nth term	$a + (n - 1)d$	$a \cdot r^{n-1}$

Sum to n terms	$\frac{n}{2}[2a+(n-1)d]$	$a(r^n-1)/(r-1)$
Sum to infinity	Always diverges (unless all terms zero)	Converges iff $\lvert r\rvert<1$, with $S_\infty = a/(1-r)$
Test if given	Successive differences constant	Successive ratios constant
Example	$3, 7, 11, 15, \ldots$	$3, 6, 12, 24, \ldots$

Worked Example 4 — GP with known sum

A GP has 5 terms with first term 2 and sum 62. Find the common ratio ($r > 0$).

```
S_n = a(r^n - 1)/(r - 1):
   62 = 2(r⁵ - 1)/(r - 1).
⇒ 31 = (r⁵ - 1)/(r - 1) = 1 + r + r² + r³ + r⁴.
Try r = 2:  1 + 2 + 4 + 8 + 16 = 31 ✓.
∴ r = 2.     // common ratio
Verify series: 2, 4, 8, 16, 32. Sum = 62 ✓.
```

Worked Example 5 — Sum to infinity

A convergent GP has sum to infinity 8 and first term 6. Find the common ratio, and write down the first four terms.

```
S∞ = a/(1 - r):
   8 = 6/(1 - r)  ⇒  1 - r = 6/8 = 3/4  ⇒  r = 1/4.
Check convergence:  |r| = 1/4 < 1 ✓ — valid.
First four terms:  6, 6·(1/4) = 3/2, 3/2·(1/4) = 3/8, 3/8·(1/4) = 3/32.
Sum check:  6 + 1.5 + 0.375 + 0.09375 + ...  ≈ 8 ✓.
```

Applications of GPs

- **Compound interest.** £1000 invested at 5% per year grows as $1000, 1000(1.05), 1000(1.05)^2, \ldots$ — a GP with $r = 1.05$.
- **Radioactive decay.** Mass remaining after n half-lives is $M_0, M_0/2, M_0/4, \ldots$ — a GP with $r = 1/2$.
- **Repeating decimals as fractions.** $0.\overline{3} = 0.3 + 0.03 + 0.003 + \cdots = \frac{0.3}{1-0.1} = \frac{1}{3}$.
- **Bouncing ball.** A ball dropped from h metres rebounds to a fraction r of its previous height, so total distance travelled is a converging GP sum.

4. Sigma Notation

Sigma notation compresses long sums into a compact symbol. The Greek capital sigma Σ means "add up".

$$\sum_{k=1}^{n} f(k) \;=\; f(1) + f(2) + f(3) + \cdots + f(n).$$

Here k is the *dummy index* (any letter will do — i, j, r are common alternatives). The lower limit is the starting value; the upper limit is the last value (inclusive). Changing the letter does not change the sum.

Standard Sums

Sum	Closed form	Used for
$\sum_{k=1}^{n} 1$	n	Counting
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$	AP sum (natural numbers)
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$	Sum of squares — used in Pearson correlation, moment of inertia
$\sum_{k=1}^{n} k^3$	$\left[\frac{n(n+1)}{2}\right]^2$	Sum of cubes — equals $(\sum k)^2$, a surprising identity

Linearity of Sigma

Sums distribute over addition and scalar multiplication:

$$\sum_{k=1}^{n}[\,\alpha f(k)+\beta g(k)\,]=\alpha\sum_{k=1}^{n}f(k)+\beta\sum_{k=1}^{n}g(k).$$

This is used to split a complicated sum into pieces you can evaluate using standard results.

Changing Limits

To compute $\sum_{k=m}^{n}$ where $m>1$, subtract:

$$\sum_{k=m}^{n}f(k)=\sum_{k=1}^{n}f(k)-\sum_{k=1}^{m-1}f(k).$$

Worked Example 6 — Using standard sum results

Evaluate $\displaystyle\sum_{k=1}^{20}(3k^2-2k+1)$.

```
1.  Split using linearity:
2.     Σ (3k² − 2k + 1) = 3 Σ k² − 2 Σ k + Σ 1.
3.  Apply standard results with n = 20:
4.     Σ k²  = 20·21·41/6 = 2870.
5.     Σ k   = 20·21/2    = 210.
6.     Σ 1   = 20.
7.  Combine:
8.     3(2870) − 2(210) + 20
9.     = 8610 − 420 + 20
10.    = 8210.
11. ∴ Σ_{k=1..20} (3k² − 2k + 1) = 8210.
```

Exam tip — watch the index start. $\sum_{k=0}^{n}$ has $n+1$ terms, not n. $\sum_{k=1}^{n}$ has n terms. A frequent slip is reading a GP problem as "five terms starting with $k=0$" and using $n=5$ in the formula

when the correct value is $n = 5$ with $a = a_0$, but still *five* terms. Count the terms explicitly before plugging in.

5. Binomial Expansion — Positive Integer Index

The binomial theorem gives an expansion of $(1+x)^n$ when n is a non-negative integer. The coefficients are the *binomial coefficients*, appearing in Pascal's triangle.

Binomial coefficient. For non-negative integers n and k with $0 \le k \le n$, \[\binom{n}{k} = \;^nC_k = \dfrac{n!}{k!\,(n-k)!}, \] read "n choose k". It counts the number of ways to choose k objects from n distinguishable objects and is always a positive integer.

Binomial Theorem (Positive Integer n)

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n.$$

For a general binomial $(a+b)^n$:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$

For a positive integer n the expansion has exactly $n+1$ terms and *terminates*. No convergence condition is needed: the expansion is a finite polynomial identity, true for all real x.

Pascal's Triangle (first six rows)

n	Row of coefficients $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$	Sum of row = 2^n
0	1	1
1	1, 1	2
2	1, 2, 1	4

3	1, 3, 3, 1	8
4	1, 4, 6, 4, 1	16
5	1, 5, 10, 10, 5, 1	32
6	1, 6, 15, 20, 15, 6, 1	64

Each entry is the sum of the two entries diagonally above: $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ — this is *Pascal's rule*.

Exam tip — finding a specific term. In $(a+b)^n$ the term containing b^k is $\binom{n}{k}a^{n-k}b^k$. For example, the x^3 term in $(1+2x)^6$ comes from $k=3$: coefficient $= \binom{6}{3} \cdot 1^3 \cdot 2^3 = 20 \cdot 8 = 160$, so the term is $160x^3$. You rarely need the whole expansion — just isolate the term you want.

Worked Example 7 — Binomial expansion of $(1 + 2x)^6$

Expand $(1+2x)^6$ up to and including the term in x^3.

```
1.  Use (1 + u)^6 = 1 + 6u + 15u² + 20u³ + ...   with u = 2x.
2.  Term-by-term:
3.     1
4.     6 · (2x)        = 12x
5.     15 · (2x)²      = 15 · 4x²   = 60x²
6.     20 · (2x)³      = 20 · 8x³   = 160x³.
7.  ∴ (1 + 2x)⁶ = 1 + 12x + 60x² + 160x³ + ...     (terms in x⁴, x⁵, x⁶ omitted).

// Full expansion (for interest, not required):
// 1 + 12x + 60x² + 160x³ + 240x⁴ + 192x⁵ + 64x⁶.
```

Worked Example 8 — Specific term in $(3 - x)^8$

Find the coefficient of x^4 in the expansion of $(3-x)^8$.

```
General term:  C(8, k) · 3^(8-k) · (-x)^k  =  C(8, k) · 3^(8-k) · (-1)^k · x^
Set k = 4:
   coef = C(8, 4) · 3⁴ · (-1)⁴
        = 70 · 81 · 1
        = 5670.
∴ coefficient of x⁴ is 5670.
```

6. Binomial Expansion — Rational / Negative Index

For a real (not necessarily integer) exponent n, the binomial series still has a meaning, but as an *infinite* series that only converges for $|x| < 1$.

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots, \qquad |x| < 1.$$

Notice two differences from the integer case:

- The expansion is infinite — it does not terminate because $n(n-1)(n-2)\cdots(n-k+1)$ is never zero when n is not a non-negative integer.
- It is only valid for $|x| < 1$. Outside this range the series diverges and gives nonsense.

Generalised binomial coefficient. For any real n and non-negative integer k, \[\binom{n}{k} = \dfrac{n(n-1)(n-2)\cdots(n-k+1)}{k!}. \] When n is a non-negative integer and $k > n$ one of the factors in the numerator is zero, so $\binom{n}{k} = 0$, recovering the terminating case.

Expanding $(a+bx)^n$

When the constant inside the brackets is not 1, factor it out first:

$$(a+bx)^n = a^n\left(1 + \frac{bx}{a}\right)^n, \qquad \text{valid for } \left|\frac{bx}{a}\right| < 1, \text{ i.e. } |x| < |a/b|.$$

Then expand $(1+u)^n$ with $u = bx/a$ and multiply through by a^n.

Validity Range Summary

Form	Expansion	Valid for
$(1+x)^n$, n positive integer	Finite polynomial ($n+1$ terms)	All real x
$(1+x)^n$, n negative or fractional	Infinite series	$\lvert x\rvert < 1$
$(a+bx)^n$, n negative or fractional	$a^n(1+bx/a)^n$, infinite series	$\lvert bx/a\rvert < 1$, i.e. $\lvert x\rvert < \lvert a/b\rvert$
$(1-x)^n$, n negative or fractional	Replace x with $-x$ in standard expansion	$\lvert x\rvert < 1$

Exam tip — always state the validity range. OCR mark schemes award an explicit mark (often one of three or four) for stating the correct range of x for which the expansion is valid. For $(1+3x)^{-2}$, valid for $\lvert 3x\rvert < 1$, i.e. $\lvert x\rvert < 1/3$. Writing just "valid for $\lvert x\rvert < 1$" is wrong here — the coefficient inside matters.

Worked Example 9 — First four terms of $(1 + 3x)^{-2}$

Find the first four terms of the expansion of $(1+3x)^{-2}$ in ascending powers of x, and state the range of x for which it is valid.

```
General expansion:
   (1 + u)^n = 1 + nu + n(n-1)/2! · u² + n(n-1)(n-2)/3! · u³ + ...
Here n = -2, u = 3x.
Compute coefficients:
   n           =  -2
   n(n-1)/2!   =  (-2)(-3)/2    =  3
   n(n-1)(n-2)/3! = (-2)(-3)(-4)/6 = -4.
Terms:
   1
   (-2)(3x)           = -6x
   3 · (3x)²          = 3 · 9x²     = 27x²
   (-4) · (3x)³       = -4 · 27x³   = -108x³.
∴ (1 + 3x)⁻² = 1 - 6x + 27x² - 108x³ + ...
```

```
14. Validity:  |3x| < 1   ⇒   |x| < 1/3.
```

7. Approximation Using the Binomial

The binomial series lets us approximate powers of numbers close to 1. If x is small the first few terms of $(1+x)^n$ give a good estimate.

Worked Example 10 — Approximating √1.02

Use a binomial expansion up to the x^2 term to approximate $\sqrt{1.02}$, and estimate the error.

```
1.  Write √1.02 = (1 + 0.02)^(1/2).
2.  Expansion with n = 1/2, u = 0.02:
3.     (1 + u)^(1/2) = 1 + (1/2)u + (1/2)(-1/2)/2! · u² +  ...
4.                   = 1 + u/2 - u²/8 +  ...
5.  Substitute u = 0.02:
6.     ≈ 1 + 0.01 - (0.0004)/8
7.     = 1 + 0.01 - 0.00005
8.     = 1.00995.

9.  Calculator value:  √1.02 = 1.00995049 ...
10. Error ≈ 5 × 10⁻⁸ - extremely accurate because u is small.

// The next term (u³ term) has coefficient (1/2)(-1/2)(-3/2)/6 = 1/16,
// giving u³/16 ≈ (8 × 10⁻⁶)/16 = 5 × 10⁻⁷, consistent with observed error.
```

Exam tip — how many terms for a target accuracy? Each successive term in the binomial series shrinks roughly by a factor of x (when $|x|$ is small). So if you want four-decimal-place accuracy and $|x| = 0.02$, two terms past the constant usually suffice. Rule of thumb: the first omitted term is the dominant error — estimate it and check it is below your tolerance.

8. Recurrence Relations — Convergence to a Limit

Some recurrence relations $a_{n+1} = g(a_n)$ produce sequences that settle down to a limit L. If $a_n \to L$ and g is continuous, then taking limits of both sides of $a_{n+1} = g(a_n)$ gives $L = g(L)$ — the limit must be a *fixed point* of g.

$$\text{If } a_n \to L \text{ as } n \to \infty, \text{ then } a_{n+1} \to L \text{ too, so } L = g(L).$$

Solving $L = g(L)$ locates the candidate limits. Not every fixed point is attracting; whether the sequence actually converges depends on the starting value and on $|g'(L)|$ (a detail beyond the specification), but candidate limits are found by this fixed-point equation.

Worked Example 11 — Convergent recurrence

Let $a_{n+1} = \frac{1}{2}(a_n + 4)$ with $a_1 = 0$. Compute $a_1, \ldots, a_5$, and find the limit L.

```
1.  Compute terms:
2.     a₁ = 0
3.     a₂ = (0 + 4)/2    = 2
4.     a₃ = (2 + 4)/2    = 3
5.     a₄ = (3 + 4)/2    = 3.5
6.     a₅ = (3.5 + 4)/2  = 3.75
7.     a₆ = (3.75 + 4)/2 = 3.875
8.  Terms approach 4 from below.

// Fixed point equation: assume limit L exists and let n → ∞.
9.  L = (L + 4)/2
10. 2L = L + 4
11. L = 4.          ∴ aₙ → 4 as n → ∞.

// Distance to limit halves each step:  |aₙ − 4| → 4, 2, 1, 0.5, 0.25, ...
// geometric decrease with ratio 1/2, confirming convergence.
```

Worked Example 12 — Recurrence giving an AP

A sequence satisfies $u_{n+1} = u_n + 5$ with $u_1 = 3$. Find a closed-form expression for u_n and hence u_{20} and $\sum_{k=1}^{20} u_k$.

```
1.  Recurrence u_{n+1} − u_n = 5 identifies an AP with
        first term a = 3 and common difference d = 5.
2.  Closed form:  u_n = a + (n − 1)d = 3 + 5(n − 1) = 5n − 2.
3.  Check:  u₁ = 3 ✓, u₂ = 8 ✓, u₃ = 13 ✓.
4.  u₂₀ = 5·20 − 2 = 98.
5.  Sum of first 20 terms:
6.     S₂₀ = 20/2 · (u₁ + u₂₀) = 10 · (3 + 98) = 10 · 101 = 1010.
7.  Check with other formula:
8.     S₂₀ = 20/2 · [2·3 + 19·5] = 10 · [6 + 95] = 10 · 101 = 1010 ✓.
```

9. Worked Exam-Style Problems — Combined Techniques

Worked Example 13 — Repeating decimal as a fraction

Express the recurring decimal $0.\overline{27} = 0.272727\ldots$ as a fraction in lowest terms using an infinite GP.

```
1.  Write as a sum:
2.     0.272727 ...  = 0.27 + 0.0027 + 0.000027 +  ...
3.  This is a GP with:
4.     first term a = 0.27 = 27/100
5.     common ratio r = 0.01 = 1/100 (each term is 1/100 of the previous).
6.  |r| = 0.01 < 1, so the series converges.
7.  Sum to infinity:
8.     S∞ = a/(1 − r) = (27/100) / (99/100) = 27/99.
9.  Simplify: gcd(27, 99) = 9, so 27/99 = 3/11.
10. ∴ 0.272727 ...  = 3/11.
11. Check:  3 ÷ 11 = 0.272727 ...  ✓.
```

Worked Example 14 — Compound interest

£2000 is deposited at 4% compound interest per year. Find the value after 10 years and the total interest earned.

```
After n years balance = 2000 · (1.04)^n — a GP with a = 2000, r = 1.04.
After 10 years:
   V = 2000 · (1.04)^10.
(1.04)^10 — expand as (1 + 0.04)^10 using binomial (n = 10, positive integer)
   = 1 + 10(0.04) + 45(0.04)² + 120(0.04)³ + ... ≈ 1.48024.
   (Calculator value: 1.48024428 ... )
V ≈ 2000 × 1.48024 ≈ £2960.49.
Interest earned ≈ 2960.49 − 2000 = £960.49.
```

Worked Example 15 — Wage calculation (AP application)

An employee earns £22 000 in year 1 and receives a £900 annual pay rise. Find total earnings over a 25-year career.

```
AP with a = 22000, d = 900, n = 25.
Salary in year n:  a_n = 22000 + 900(n − 1).
Year 25 salary:  22000 + 900·24 = 22000 + 21600 = £43600.
Sum:
   S₂₅ = 25/2 · [2·22000 + 24·900]
        = 25/2 · [44000 + 21600]
        = 25/2 · 65600
        = 25 · 32800
        = £820 000.

// Alternatively:  S₂₅ = 25/2 · (22000 + 43600) = 12.5 · 65600 = 820000. ✓
```

10. Consolidated Summary and Exam Strategy

Chapter-wide checklist for Sequences & Series.

- **Identify the sequence type first.** Common difference ⇒ AP; common ratio ⇒ GP; neither ⇒ use recurrence or general formulae.
- **AP:** $a_n = a + (n-1)d$**,** $S_n = \frac{n}{2}(a+l) = \frac{n}{2}[2a + (n-1)d]$**.**
- **GP:** $a_n = ar^{n-1}$**,** $S_n = a(r^n - 1)/(r-1)$**. Sum to infinity requires** $|r| < 1$**.**
- **Sigma notation:** memorise $\Sigma k, \Sigma k^2, \Sigma k^3$; use linearity to split complicated sums.
- **Binomial (positive integer** n**):** finite expansion with $n+1$ terms, valid for all real x; coefficients from Pascal's triangle.
- **Binomial (rational/negative** n**):** infinite series, valid for $|x| < 1$; for $(a+bx)^n$ factor out a^n first and validity becomes $|x| < |a/b|$.
- **Recurrence limits:** assume limit L exists, substitute $a_{n+1} = a_n = L$, solve $L = g(L)$.
- **State validity conditions** explicitly for infinite GP and non-integer binomial — OCR awards a mark for this.

Typical OCR examiner complaints (from mark reports).

- Using $S_\infty = a/(1-r)$ without checking or stating $|r| < 1$.
- Omitting the validity range $|x| < |a/b|$ for the expansion of $(a+bx)^n$.
- Arithmetic slips in $n(n-1)(n-2)/3!$ when n is negative — sign errors abound.
- Wrong number of terms in a sigma sum (index starts at 0 vs 1).
- Forgetting to factor a^n out of $(a+bx)^n$ before applying the series formula.
- Writing $(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \ldots$ while missing the $2!$ in the denominator of subsequent terms' factorials.
- Failing to verify a guessed explicit formula on multiple terms, not just the first.
- Mixing up AP and GP formulae under exam pressure — confirm the structure before reaching for a formula.

Formula snapshot (memorise).

- AP nth term: $a_n = a + (n-1)d$.
- AP sum: $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l)$.

- GP nth term: $a_n = a\,r^{n-1}$.
- GP sum: $S_n = a(r^n - 1)/(r - 1),\ r \neq 1$.
- GP sum to infinity: $S_\infty = a/(1 - r),\ |r| < 1$.
- Standard sigma sums: $\Sigma k = n(n+1)/2,\ \Sigma k^2 = n(n+1)(2n+1)/6,\ \Sigma k^3 = [n(n+1)/2]^2$.
- Binomial (integer n): $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$.
- Binomial (general n): $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots,\ |x| < 1$.
- Fixed point of recurrence: $L = g(L)$ if the limit exists.